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SUPERSEDING
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MILITARY SPECIFICATION

COMPUTER, BALLISTICS, M14A1

1. SCOPE

1.1 This specification covers one type of electromechanical computer known as Ballistics Computer, M14A1 of which the range servo amplifier is a separate component part. The computer transmits superelevation data to the fire control system of the 120MM Gun Tank M103A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL-S-16892 — Synchros, 400 Cycle.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-202 — Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

F10511327 — Computer, Ballistics, M14A1.

PACKAGING DATA SHEET

MIL-P-14232/P8602750 — Packaging of Computer Ballistics M14 and M14A1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1220

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3. REQUIREMENTS

3.1 Materials. The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication. The Computer Ballistics M14A1 shall be manufactured in accordance with Drawing F10511327 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Electrical and electronic assemblies.

(e) Part identification and marking.

(f) Workmanship.

3.4 Vibration. The computer shall be vibrated in a vertical plane at an amplitude of 1/16 inch (1/8 inch total excursion) at a constant frequency of 30 cycles per second for a duration of 5 minutes plus or minus 15 seconds. Subsequent to vibration, the computer shall show no evidence of loose or damaged parts and shall meet all the requirements of this specification.

3.5 Superelevation counter output accuracy. With the computer having a constant load of between 90 and 110 ounce inches applied to the output shaft for periscope drive-mount Drawing C8602819 the ammunition selector on the AP–T M358 Cam, the range correction knob set to zero, and the mechanical calibration shaft set at 1500 meters, actuate the superelevation hand crank to match the outer index with the inner index within half line width by approaching the inner index from clockwise and counterclockwise directions without overtravel. Establish the position of the output shaft for the periscope drive-mount to the 1500 meters check point by splitting the backlash error obtained from the clockwise and counterclockwise positions above. Superelevation counter accuracy shall be checked from the 1500 meters checkpoint by determining if the rotation of the output shaft for the periscope drive-mount is within the tolerance prescribed in Table I for the specified superelevation counter readings.

TABLE I. *Superelevation Counter Output Accuracy From the 1500 Meter Check Point for the AP–T, M358 Cam (D8602528)*

Position of output shaft for Drive Mount — M9	239° 46" CCW	1500 meter starting point 0	2R 278° 38' CW	4R 62° 38' CW
Counter reading in mils	3.8	7.13	21.0	28.0
Tolerance on output shaft position	±8°	±3°	±8°	±8°

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3.6 Limits of shaft rotations.

3.6.1 *Mechanical calibration shaft.* The mechanical stops shall limit shaft travel as specified on Drawing F8602532.

3.6.1.1 *Electrical stops.* The electrical stops shall limit rotation of the mechanical calibration shaft as specified on Drawing F10511328, Sheet 7, and shall prevent the servo input motor from driving against the mechanical stops.

3.6.2 *Output shafts.* The output shafts shall continually rotate during actuation of the hand crank from minus 1.5 plus or minus 0.1 mil to plus 68.5 plus or minus 0.5 mil as read on the superelevation counter.

3.7 **Manual operation, input-output.** Using the AP–T, M358 Cam, the mechanical calibration shaft shall be positioned to each of the ranges as indicated in Table VI. Rotation of the handcrank in each direction to match the outer and inner indexes within plus or minus one half line width shall produce rotation of the output shaft for drive-mount to correspond to the values indicated.

3.7.1 The procedure of 3.7 shall be repeated for the 30 caliber BR–M1917, the HE–T, M356 and the HEAT–T, M469 Cams.

3.8 **Manual operation, range scale-output.** Using the AP–T, M358 Cam only, alinement of the outer index to the range scale by rotation of the handcrank in each direction shall produce rotation of the output shaft for drive-mount to correspond to the values of Table VI at ranges indicated therein.

3.9 **Cam sequence.** Four cams shall have been placed in the computer in the following sequency starting with the AP–T, M358 followed by the BR, M1917. The HE–T, M356 and the HEAT–T, M469 followed by two spacers. Cams used for inspection and qualification shall remain with that computer.

3.10 Stability.

3.10.1 *Input motor.* The input motor shall not visually display self-sustained oscillatory operation at any range when the computer is energized by a range input signal at voltages and frequencies indicated below:

105 V. between 360 and 400 cycles A.C. and 18 V.D.C.

125 V. between 400 and 440 cycles A.C. and 28 V.D.C.

A maximum of three changes in direction of the input motor shall be considered as satisfactory operation. These three changes shall be counted as reversals in direction of the computer inner dial.

3.10.2 *Output motor.* The output motor shall not visually display self-sustained oscillatory motion at the output shaft when the mechanical calibration shaft is rotated between minimum range and 1,500 meters with the ammo-selector set on any cam. This shall be met with the computer energized with 28 plus or minus 0.5 volts dc, the output shaft to the periscope drive-mount loaded with 20 to 30 ounce inches and minus 15 percent range correction introduced. Each time the rotation of the calibration shaft is stopped, a maximum of eight changes in direction of the output motor shall be considered as satisfactory operation. These eight changes may be counted as reversals in direction of the computer outer dial.

3.10.2.1 *Output servo sensitivity.* A load of 90 to 110 ounce inches shall be applied to the output shaft and the ammunition selector set on the AP–T, M358 Cam, the servo output motor energized with 18 plus or minus 0.5 volts dc, and the automatic followup cycle completed at any range between 4000 and 4400 meters. The number of times the contacts are required to close to advance the output shaft for periscope

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drive-mount, approximately 72° shall be determined by slowly rotating the mechanical calibration shaft in one direction. The average allowable superelevation increment for one make and break of the contacts shall not exceed 10° 45′ and the maximum shall not exceed 21° 30′. Rotation of the mechanical calibration shaft in the opposite direction shall meet the above requirements.

3.11 Time of response.

3.11.1 *Input motor.* The total elapsed time for the inner index to stabilize at the indicated range on the range scale after the input motor is energized shall not exceed the values in table II at voltage indicated.

TABLE II. *Time of Response — Input Motor*

Range input signal transmitter	Inner index	Voltage ±0.5 V DC	Elapsed time for inner index to stabilize (seconds)
4400 meters	500 meters	28	3
500 meters	4400 meters	28	3
500 meters	4400 meters	18	4
4400 meters	500 meters	18	4

3.11.2 *Output motor.* With a load of 90 to 110 ounce inches applied to the output shaft for the drive-mount, the total elapsed time for the outer index to stabilize in coincidence with the inner index, after the output motor is energized when the ammunition selector is set on HE–T, M356 Cam, shall not exceed the values in Table III at the voltage indicated.

TABLE III. *Time for Response — Output Motor*

Position of outer index	Position of inner index	Voltage ±0.5 V DC	Total elapsed time (seconds)
4400 meters	500 meters	28	4
500 meters	4400 meters	28	4
500 meters	4400 meters	18	8
4400 meters	500 meters	18	8

3.12 Circuit breaker.

3.12.1 *Input motor.*

3.12.1.1 *Motor stalled.* The circuit breaker shall open n 10 seconds or less when the mechanical calibration shaft is immobilized and a range signal is applied to the input synchro transmitter with the motor energized with 28 plus 0.5 volts.

3.12.1.2 *Continuous operation.* The cir-

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cuit breaker shall remain in the closed position when the motor is energized with 28 plus 0.5 volts and subjected to 25 reversals over a period of 45 seconds.

3.12.2 *Output motor.*

3.12.2.1 *Motor stalled.* With the output shaft immobilized, the outer and inner indexes separated by at least 200 meters, and the motor energized with 28 plus 0.5 volts, the circuit breaker shall open in 90 seconds or less.

Note: The above requirement shall be performed when the motor is cold (approximately normal room temperature).

3.12.2.2 *Continuous operation.* With a load of 90 to 110 ounce inches applied to the output shaft for the drive-mount the circuit breaker shall remain in the closed position when the mechanical calibration shaft is oscillated at a rate of four to four and one-half revolutions per second for 45 seconds over a spread of 1000 meters at ranges between 1200 and 2800 meters. This requirement shall be met with the ammo selector set on the AP–T, M358 Cam and the motor energized with 28 plus 0.5 volts dc.

3.13 **Operability.** The operating torques shall not exceed the values of Table IV at temperatures indicated therein.

TABLE IV. *Operating torque values*

Control means	60° to 90°F., and 150°F.	—40°F.	Type of operation
Correction knob	34 oz. in.	45 oz. in.	Rotation.
Correction knob	20 lb.	24 lb.	Pull out force.
Ammo selector	60 lb. in.	80 lb. in.	Maximum lift of follower.
Ammo selector	10 lb.	20 lb.	To translate follower.
Handcrank	6 lb. in.	20 lb. in.	Rotate while engaged.
Handcrank	12 lb.	20 lb.	To hold engaged.
Circuit breakers	Electrically and mechanically operable.	Electrically and mechanically operable.	On—Off position.
Interrupter switch	Operable	Operable	In accordance with 3.19.

3.14 *Computer accuracy.*

3.14.1 *Electrical input.* The range signal from the input transmitter shall produce angular rotation of the mechanical calibration shaft from the 1500 meters index in accordance with the values in table V.

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TABLE V. *Range Input Transmitter Rotation As Established By Exact Positioning Of Calibration Shaft*

Position of mechanical calibration shaft.	11–R 68.5° CW ±1°	4–R 46.8° CW ±1°	*0° check-point.	2–R 334.9° CCW ±1°	7–R 21.7° CCW ±1°	9–R 356.6° CCW ±1°	10–R 346.1° CCW ±1°
Position of range input transmitter rotor.	78° 41' CW	29° 2' CW	0°	20° 36' CCW	49° 39' CCW	70° 15' CCW	77° 4' CCW
Equivalent of range in meters.	500	1000	1500	2000	3000	4000	4400

* Check point as established on mechanical calibration shaft in accordance with Drawing F10511328.

3.14.2 *Output, electromechanical.* The requirements shall be met with the output motor energized with 28 plus or minus 0.5 volts then with 18 plus or minus 0.5 volts and a load of 90 to 110 ounce inches applied to shaft for periscope drive-mount.

3.14.2.1 *Mechanical.* The position of the output shaft for the periscope drive-mount as determined from the position of the mechanical calibration shaft shall be in accordance with the values in table VI.

3.14.2.2 *Electrical.* The position of the fine and coarse synchros as derived from the position of the mechanical calibration shaft shall be in accordance with the values in table VIII.

TABLE VI. *Position of Shaft for Periscope Drive-Mount M9 as derived from position of Mechanical Calibration Shaft, Based on 1500 Meter Reference Range for AP–T, M358 Cam (D8602528)*

Range in meters	500	1000	1500	2000	3000	4000	4400
Position of mechanical calibration shaft	11–R 68.5° CW	4–R 46.8° CW	0° check point	2–R 334.9° CCW	7–R 21.7° CCW	9–R 356.6° CCW	10–R 346.1° CCW
AP–T, M358	———	180° 43' CCW	0°	190° 48' CW	1–R 247° 41'	3–R 0° 0'	3–R 207° 22'
Drive-Mount M9	———	*4.62	*7.13	*9.78	*15.57	*22.13	*25.01
BR–M1917 Drive–Cal–30 Mount M9	136° 48' CCW	2–R 27° 22' CW	———	———	———	———	———
	*5.23	*17.51					
Tolerance on output shaft to Drive-Mount M9 for BR M1917 Cam Only	±10° 48	±18°	———	———	———	———	———
	*0.15	*0.25					

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TABLE VI.—Continued.

Range in meters	500	1000	1500	2000	3000	4000	4400
HE–T, M356 Drive-Mount M9	195° 50' CCW *4.41	141° 50' CW *9.10	1–R 140° 24' CW *14.08	2–R 160° 34' CW *19.36	4–R 270° 43' CW *30.89	7–R 134° 38' CW *44.00	8–R 191° 31' CW *49.79
HEAT–T, M469 Drive–Mount M9	1–R 10° 48' CCW *1.98	187°55' CCW *4.52	31° 41' CW *7.57	295° 55' CW *11.24	2–R 324° 43' CW *21.64	6–R 194° 24' CW *39.83	8–R 257° 46' CW *50.71
Tolerance on output shaft to Drive-Mount M9 for M358, M356 and M469 cams.	±10° 48' *0.15	±10° 48' *0.15	±7° 12' *0.10	±10° 48' *0.15	±18° *0.25	±28° 48' *0.40	±39° 36' *0.55

*Corresponding nominal superelevation value in mils (informational only).

TABLE VII. *Position of Rotation of Rotors of the Computer Control Transmitters as Determined by Position of External Receivers at Null Position, based on 1500 Meter Reference Range for AP–T, M358 Cam (D8602428)*

Range in meters		500	1000	1500	2000	3000	4000	4400
Position of mechanical calibration shaft		11–R 68.5° CW	4–R 46.8° CW	0° Check point	2–R 334.9° CCW	7–R 21.7° CCW	9–R 356.6° CCW	10–R 346.1° CCW
Cam	Synchro F	———	36° 8' CW	0°	38° 10' CCW	121° 32' CCW	216° 32' CCW	257° 28' CCW
AP–T, M358	*C	———	0° 42' CW	0° C	0° 45' CCW	2° 22' CCW	4° 13' CCW	5° 2" CCW
BR–M1917 Cal. 30	F	27° 22' CW	129° 28' CCW	———	———	———	———	———
	*C	0° 32' CW	2° 32' CCW	———	———	———	———	———
Tolerance for BR–M1917 Cam Only		±2° 9	±3° 35'	———	———	———	———	———

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TABLE VII.—*Continued*

Range in meters		500	1000	1500	2000	3000	4000	4400
HE–T, M356	F	39° 10' CW	28° 22' CCW	100° 5' CCW	176° 7' CCW	342° 9' CCW	1–R 170° 56' CCW	1–R 254° 18' CCW
	*C	0° 46' CW	0° 33' CCW	1° 57' CCW	3° 26' CCW	6° 41' CCW	10° 22' CCW	12° 0' CCW
HEAT–T, M469	F	74° 10' CW	37° 35' CW	6° 20' CCW	59° 11' CCW	208° 57' CCW	1–R 110° 53' CCW	1–R 267° 33' CCW
	*C	1° 27' CW	0° 44' CW	0° 7' CCW	1° 9' CCW	4° 5' CCW	9° 11' CCW	12° 15' CCW
Tolerance (Fine only) for Cam M358, M356 and M469	F	±2° 9' *0.15 mil	±2° 9' *0.15	±1° 26' *0.01	±2° 9' *0.15	±3° 35' *.25	±5° 44' *0.40	±7° 53' *0.55 mil

Note. C = Coarse synchro. F = Fine synchro.

* Information only.

3.15 Backlash. Backlash shall not exceed 1° at any setting at the output shafts to the periscope cant corrector, superelevation generator and the range-finder cant corrector when shaft for periscope drive-mount is secured.

3.16 Environmental requirements.

3.16.1 *Storage temperatures.* The computer shall be exposed to ambient temperatures of minus 80°F. and plus 160°F. until thermal equilibrium is reached. After thermal equilibrium has been re-established at room temperature, the computer shall show no evidence of physical failure such as casting failure or glass breakage, and shall also meet the requirements of 3.17.

3.16.2 *Operational temperatures.* The computer shall be exposed to ambient temperatures of minus 40°F. and plus 150°F. until thermal equilibrium is reached at each temperature. The computer shall then be operated with 18 plus or minus 0.5 volts, utilizing the HE–T, M356 Cam with a load of 90 to 110 ounce inches applied to drive-mount shaft. With range data introduced through a range input signal transmitter, the computer shall be inspected at each temperature to the following requirements.

(a) Applicable requirements of 3.13

(b) The outer index shall follow the inner index within one-half a graduation width when the inner index is advanced over its maximum excursion of the range scale in each direction.

(c) Requirements of 3.10 at 150°F. only.

3.16.2.1 Immediately upon completion of the environmental requirements of 3.16 the

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interior of the computer shall be dried by clean, dry, warm air of approximately 100° to 140°F.

3.16.3 *Water resistance.* The interior of the computer shall show no evidence of water penetration after being subjected to a simulated rain fall as required by the rain test of Specification MIL–F–13926, Procedure I. A time duration of 15 minutes shall be substituted for 2 hours.

3.17 **Post environmental check.** Computers that have been subjected to the environmental requirements of 3.16 shall upon return to room temperature (60° to 90°F.) meet the requirements of 3.10, 3.14.1 and 3.14.2.2.

3.18 **Illumination.** The internal lighting device of the computer shall provide adequate illumination to easily identify the inner and outer indexes of the range dial and the ammunition index when observed in a darkened area.

3.19 *Interrupter switches.* When the computer hand crank is engaged, the interrupter switch shall open the output motor circuit. Releasing the hand crank shall cause the output motor circuit to be closed. Actuation of the ammo selector handle shall cause the ammo selector interrupter switch to open the output motor circuit. Re-setting of the handle shall cause the switch to close the circuit.

3.20 The following electrical requirements may be determined any time prior to final acceptance of the computer.

3.20.1 *High potential.* Insulation breakdown shall not occur between isolated circuits and between electrical circuits and body or frame when 500 volts (RMS) is applied for a period of between 5 seconds and 10 seconds.

3.20.2 *Insulation resistance.* Following the high potential test, insulation resistance between ungrounded electrical circuits and the body or frame and between isolated circuits shall not be less than 5 megohms.

3.20.3 *Continuity.* All electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

3.20.4 *Electrical zero.* Synchros shall be electrically zeroed in accordance with Drawings D8602859 and F10511328 Sheet 7.

3.21 **Identification.** Each range servo amplifier shall be identified as a component of the computer with which it qualified by stenciling the serial number of the computer on the amplifier. Above two units shall then be packaged together.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspec-

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tion terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in Table VIII titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE VIII. *Classification of Defects*

Use Inspection Level II of Table I with Sampling Plan Table II–A of Standard MIL–STD–105

	Requirement	*Test procedure*
Critical: None.		
Major: AQL .65 Percent Defective		
101. Vibration	3.4	4.5.2
102. Insufficient accuracy superelevation counter.	3.5	4.5.6
103. Mechanical stops, not within limits	3.6.1	4.5.7
104. Electrical stops, not within limits	3.6.1.1	4.5.7.1
105. Output shafts, not within limits	3.6.2	4.5.8
106. Manual operation input-output — not within limits.	3.7	4.5.9
107. Manual operation range scale output—not within limits.	3.8	4.5.10
108. Cam sequence, operation failure	3.9	4.5.11
109. Input motor-not within limits	3.10.1	4.5.12
Major: AQL .65 Percent Defective		
110. Output motor-not within limits	3.10.2	4.5.13
111. Output servo sensitivity not within limits	3.10.2.1	4.5.14
112. Input motor, time response not within limits.	3.11.1	4.5.15

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TABLE VIII.—*Continued.*

	Requirement	*Test procedure*
113. Output motor, time response not within limits.	3.11.2	4.5.16
114. Motor stalled (input)-not within limits	3.12.1.1	4.5.17
115. Continuous operation failure	3.12.1.2	4.5.18
116. Motor stalled (output)-not within limits	3.12.2.1	4.5.19
117. Continuous operation failure	3.12.2.2	4.5.20
118. Operability-Excess torque	3.13	4.5.21
119. Electrical input-not within limits	3.14.1	4.5.22
120. Mechanical output-not within limits	3.14.2	4.5.23
121. Electrical output-not within limits	3.14.2.2	4.5.24
122. Backlash-excess insufficient	3.15	4.5.25
123. Illumination inoperable	3.18	4.5.26
124. Interrupter switches insulation breakdown	3.19	4.5.27
125. High potential insufficient	3.20.1	4.5.28
126. Insulation resistance	3.20.2	4.5.28.1
127. Lack of continuity	3.20.3	4.5.28.2
128. Improper electrical zero	3.20.4	4.5.28.3
129. Improper identification	3.21	4.5.29

Minor: None defined.

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL–STD–105.

4.3 **Control tests.** One computer shall be selected at random from each 25 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control Test	*Requirement*	*Test Procedure*
(a) Fabrication	3.2	4.5.1
(b) General Specifications	3.3	Visual, MIL–F–13926
(c) Operating temperature	3.16.1	4.5.3, 4.5.5, 4.5.2.1
(d) Storage temperature	3.16.2	4.5.3, 4.5.5
(e Water resistance	3.16.3	4.5.4

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4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted sample have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Equipment used for environmental testing.* All environmental testing equipment used to determine conformance of the environmental requirements for Vibration, Temperature, and Water Resistance shall conform to the "Test Facilities" requirements outlined in Specification MIL–F–13926.

4.4.2.3 *Test equipment.* In addition to the above mentioned requirements the following test equipment shall be used.

4.4.2.3.1 *Equipment to test functional requirements.* Final acceptance test equipment shall be capable of determining acceptable or rejectable computers for those requirements specified in 3.5 to 3.20 inclusive. The devices shall be so constructed as to permit location of the ballistics computer in its normal operating position, and shall have sufficient mediums to test the requirements listed above. The mediums noted below shall be to the accuracies stated. The medium used to view the inner and outer index shall be 3 power or better with a line width of which shall be 0.008 plus or minus 0.001. The medium used to view the superelevation counter of the ballistics computer shall be 3 power or better. The medium used to supply power for testing the requirements of stability, time of response, circuit breaker, computer accuracy and environmental requirements shall be as follows:

(a) Input voltage range 95 to 130 volts a-c maximum input current approximately 35 amperes a-c.

(b) Output voltage range 3 to 30 volts d-c continuously adjustable.

(c) Regulation accuracy plus or minus 0.25 percent from 95 to 130 volts a-c line change at any fixed load from 3 to 30 amperes d-c.

(1) Plus or minus 2 percent from 95 to 130 volts a-c line change at any fixed load from 0 to 3 amperes d-c.

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(2) Plus or minus 0.25 percent from 3 to 30 amperes d-c load change at any fixed line voltage from 95 to 130 volts a-c.

(3) Plus or minus 2 percent from 0 to 30 amperes d-c load change at any fixed line voltage from 95 to 130 volts a-c.

(d) Recovery time 0.2 seconds.

(e) Input frequency range 50 to 60 cycles per second.

(f) Load range 0 to 30 amperes.

(g) Ripple 1 percent RMS maximum of the output setting this rating applies to a nominal input frequency of 60 cycles at normal room temperature.

(h) This unit will operate at ambients from minus 40°F. to plus 150°F.

The medium used to establish pertinent positions of the mechanical calibration shaft while testing the requirements of: superelevation counter output accuracy, mechanical calibration shaft, electrical stops, manual operation input output, manual operation, range scale output, output motor, output zervo sensitivity, input motor motor stalled, output motor motor stalled, continuous operation, computer accuracy electrical input, computer accuracy mechanical, and computer accuracy electrical, shall be graduated in meters and degrees with an angular tolerance of plus or minus 5 minutes total. The medium shall incorporate a means of locking. There shall also be a method to establish amounts of rotation. The medium used to establish pertinent positions of the output shaft periscope drive-mount while testing the requirements of superelevation counter output accuracy, manual operation input output, manual operation, range scale output, output motor, output servo sensitivity, output motor motor stalled, continuous operations, computer accuracy output electro mechanical, computer accuracy mechanical, and operational temperatures, shall be graduated in degrees with an angular tolerance of plus or minus 9 minutes total. The medium shall incorporate a means of locking. There shall also be a method to establish amounts of rotation. The medium shall incorporate a method of torque, adjustable, within 20–110 ounce inches. The medium used to check backlash of output shafts to the periscope cant corrector, superelevation generator and the rangefinder cant corrector shall be graduated 360 degrees in degree increments with an angular total tolerance to be within plus or minus 6 minutes. The medium used to establish pertinent positions of the range signal while testing the requirements of: stability input motor, time of response output motor, circuit breaker continuous operation, computer accuracy electrical input, and operational temperatures, shall be graduated from 440 meters to 4800 meters within total angular tolerance plus 0.5 minutes and shall be equipped with a method of transmitting this range electrically to the ballistic computer through an amplifier. The medium used to establish pertinent positions of the superelevation output signal receiver while testing the requirements of computer accuracy electrical coarse and fine synchros, shall be graduated a full 360 degrees to the ½ degree with a total tolerance plus or minus 1 minute and shall be equipped with a method of receiving the electrical output from the ballistics computer.

4.4.2.3.2 *Equipment to test torque—correction knob rotation.* The equipment shall have a minimum range of from zero to 45 inch ounces, the accuracy of which shall be plus or minus ¼ inch ounces throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate testing of the correction knob.

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4.4.2.3.3 *Equipment to test force—correction knob pull.* The equipment shall have a minimum range of from zero to 24 pounds, the accuracy of which shall be plus or minus ¼ pound throughout the entire zone excursion, the device shall be equipped with adapter to accommodate testing of the correction knob.

4.4.2.3.4 *Equipment to test torque—ammo selector rotation.* The equipment shall have a minimum range of from zero to 80 pound inches throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the ammo selector.

4.4.2.3.5 *Equipment to test force—ammo selector pull.* The equipment shall have a minimum range of from zero to 20 pounds the accuracy of which shall be plus or minus ¼ pound throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the ammo selector.

4.4.2.3.6 *Equipment to test torque—handcrank rotation.* The equipment shall have a minimum range of from zero to 20 pound inches, the accuracy of which shall be plus or minus ¼ pound inch throughout the entire zone of excursion. The device shall be equipped with adapter to accomodate testing of the handcrank.

4.4.2.3.7 *Equipment to test force—handcrank push.* The equipment shall have a minimum range of from zero to 20 pounds the accuracy of which shall be plus or minus ¼ pound throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the handcrank.

4.5 Test methods and procedures.

4.5.1 *Fabrication.* The computer shall be subjected to a visual, tactile and dimensional inspection for conformance to the applicable drawings, and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 *Vibration.* The computer shall be clamped to a vibration table with the mounting surface of the mount horizontal and parallel to the table. The periscope drive-mount output shaft shall be supported and immobilized and shall remain immobile throughout the test. The computer shall be vibrated at the amplitude and frequency specified in 3.4. The vibration amplitude shall be measured at the extremeties of the computer as positioned on the vibration table. Subsequent to vibration, the computer shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The computer shall then be subjected to the tests specified in 4.5.2 to 4.5.29 inclusive and shall conform to the requirements specified therein.

4.5.3 *Storage and operating temperatures.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 at the temperatures specified in 3.16.1 and 3.16.2. While stabilized at each temperature extreme specified in 3.16.2 the computer shall be operated as specified and shall meet the applicable requirements therein when subjected to the tests specified in 4.5.12, 4.5.13 and 4.5.14. The controls of the computer shall meet the specified torque and force values of Table IV and applicable requirements of 3.16.2 for the indicated operating temperature using the test equipment specified in 4.4.2.3. Upon completion of the specified tests and return to standard ambient temperature (60° to 90°F.) the computer shall be inspected for damage such as casting failure or glass breakage and shall then be subjected to the tests specified in 4.5.5 and shall conform to the requirements specified.

4.5.4 *Water resistance.* This test shall be conducted in accordance with Procedure I of the rain test in Specification MIL–F–13926 for the time period specified in 3.16.3.

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Upon completion of the test, the computer cover shall be removed and the area within visually examined for evidence of water penetration or damage to determine compliance with the requirements of 3.16.3. The computer shall then be subjected to the tests specified in 4.5.5 and shall conform to the requirements specified.

4.5.5 *Post environmental check.* Subsequent to the environmental tests of 4.5.3 and 4.5.4 the computer shall be subjected to the tests specified in 4.5.12, 4.5.13, 4.5.14, 4.5.22 and 4.5.24 at standard ambient temperatures (60° to 90°F.) and shall conform to the requirements specified to determine compliance with 3.17.

4.5.6 *Superelevation counter output accuracy.* Superelevation counter output accuracy shall be inspected to the requirements of 3.5 and Table I when mounted on special test equipment specified in 4.4.2.3.1 and applying the constant load on the periscope drive-mount output shaft. A visual reading of the range scale and the inner matching dial index shall be made. The superelevation counter scale shall be read with an accessory of the special test equipment having a reticle line for reading to (0.01) Mil accuracy. The mechanical calibration shaft shall be inspected for input meters, and revolutions.

4.5.7 *Mechanical calibration shaft.* This test shall be accomplished with special testing equipment as outlined in 4.5.6 and it shall meet the requirements of 3.6.1 and applicable drawings. The shaft shall be rotated slowly when approaching the stops. The stops shall not be struck with force.

4.5.7.1 The electrical stops shall be tested with special test equipment, capable of reading the range and electrically indicating when the servo motor is not energized and meeting the requirements of 3.6.1.1 and applicable drawings.

4.5.8 *Output shafts.* Rotation of the output shafts shall be visually observed during actuation of the hand crank. The special testing equipment in 4.2.2.1 shall be used to determine compliance with the tolerance requirements of 3.6.2.

4.5.9 *Manual operation, input-output.* With the mechanical calibrated shaft affixed with a special testing equipment as outlined in 4.2.3.1 and the output shaft for the drive-mount affixed with special test equipment capable of measuring angular rotation in revolutions, rotation of the hand crank in each direction to match the outer and inner indexes when viewed through the special testing equipment provided for the test shall meet the manual operation requirements of 3.7 and 3.7.1.

4.5.10 *Manual operation, range scale-output.* This inspection shall be conducted using special testing equipment outlined in 4.4.2.3.1 for reading the range scale, and for measurement of the output drive-mount shaft by rotation. Rotation of the handcrank in each direction shall align the outer index to the range scale and produce rotation of the output drive-mount shaft to determine compliance with the requirements of 3.8.

4.5.11 *Cam sequence.* This shall be a visual inspection to insure that the sequence is correct as required by 3.9. The cams used shall be those with which the computer was qualified.

4.5.12 *Input motor.* This inspection test shall be conducted using special testing equipment, which shall be capable of supplying an electrical range signal input to the computer, and a dc power supply capable of such capacity and stability as to maintain the specified voltages within three percent during operation of the computer. The ac power supply shall be capable of maintaining the specified voltages and frequencies within ten percent during operation of the computer. A check for oscillatory operation and the permissible maximum of three

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changes in direction of the input motor shall be visually observed on the computer inner dial to meet the requirements of 3.10.1.

4.5.13 *Output motor.* With the constant specified load applied to the periscope drive-mount output shaft by special testing equipment and the range input indicated on the mechanical calibrated shaft by special testing equipment, a check for oscillatory operation and the permissible maximum of eight changes in direction of the output motor shall be visually observed on the computer outer range dial to determine compliance with 3.10.2.

4.5.14 *Output servo sensitivity.* The periscope output drive-mount shaft shall have the specified constant load applied by special testing equipment. The range shall be read from the range scale special testing equipment. The angular rotation of the periscope drive-mount shaft shall be indicated by special testing equipment when the mechanical calibrated input shaft is slowly rotated. The servo clicks shall be counted and averaged out to meet the requirements of 3.10.2.1.

4.5.15 *Input motor.* With the servo input motor circuit breaker in the off position, the special test equipment shall be capable of electrically transmitting a range signal input at the specified ranges in 3.11.1, table II. The inner index shall be positioned at the range specified. The elapsed time (after the power has been applied by resetting the circuit breakers) for the inner index to stabilize within half a line width on the range scale, when timed with standard measuring equipment shall not exceed the tolerances specified.

4.5.16 *Output motor.* With the specified constant load applied to the periscope drive-mount shaft and the inner index positioned to the specified ranges indicated in table III by special testing equipment the hand crank shall be rotated to the specified ranges for the outer index. The elapsed time after the power has been applied, by resetting the circuit breaker, for the inner index to stabilize within half a line width on the range scale, when timed with standard measuring equipment, shall meet the requirements of 3.11.2 and table III.

4.5.17 *Motor stalled.* With the mechanical calibration shaft locked by special testing equipment and the motor energized at the specified voltage, the length of time for the circuit breaker to open, shall be timed with standard measuring equipment, to determine compliance with 3.12.1.1.

4.5.18 *Continuous operation.* The special test equipment specified in 4.4.2.3.1 shall be capable of electrically transmitting range to the computer subject the motor to the required reversals. Standard measuring equipment shall be used to time the continuous operation as specified in 3.12.1.2.

4.5.19 *Motor stalled.* With the periscope drive-mount output shaft locked by special testing equipment and the motor energized at the specified voltage, the length of time for the circuit breaker to open shall be timed with standard measuring equipment, to meet the requirements of 3.12.2.1.

4.5.20 *Continuous operation.* The specified load on the output shaft for the periscope drive-mount shall be applied by special testing equipment. The mechanical calibrated input shaft shall be oscillated at the specified rate, over the required spread of ranges by use of standard measuring equipment to meet the requirements of 3.12.2.2.

4.5.21 *Operability.* The torques and forces required to actuate the controls of the computer shall not exceed the applicable values specified in Table IV while at the ambient temperature of plus 60°F to plus 90°F. These tests shall be conducted with special testing equipment and standard measuring equipment capable of measuring the required torques and forces in 3.13.

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4.5.22 *Electrical input.* This inspection test shall be conducted utilizing the special test equipment used in 4.5.15. The range input signal shall be used for checking the angular rotation of the mechanical calibration input shaft specified in Table V with the periscope drive-mount output shaft having the constant specified load applied and the output motor energized with the required voltages as specified in 3.14.2 to meet the requirements of 3.14.1. Tests 4.5.23 and 4.5.24 shall meet the requirements of 3.14.1.

4.5.23 *Mechanical — output.* When the mechanical calibration input shaft is positioned to the specified values by special testing equipment the position of the periscope drve-mount output shaft, when indicated by special testing equipment shall conform to the values and tolerances of Table VI as specified in 3.14.2.

4.5.24 *Electrical — output.* The course and fine synchro position shall be indicated by special testing equipment capable of reading angular rotation in degrees to the specified values and tolerances when the mechanical calibration output shaft is positioned in accordance with Table VII and 3.14.2.2.

4.5.25 *Backlash.* The backlash shall be inspected on special testing equipment. Visual observation of the output shafts shall be made and must meet the angular requirements of 3.15.

4.5.26 *Illumination.* Illumination shall be visually observed in a darkened area to meet the requirements of 3.18.

4.5.27 *Interrupter switches.* When the computer handcrank is engaged and the ammo selector handle is actuated as specified, the results shall be visually observed to meet the requirements of 3.19.

4.5.28 *High potention.* This test shall be conducted in accordance with Standard MIL–STD–202 and shall meet the requirements of 3.20.1.

4.5.28.1 *Insulation resistance.* Immediately following the high potential test, this test shall be conducted in accordance with standard MIL–STD–202 and shall meet the requirements of 3.20.2.

4.5.28.2 *Continuity.* This test shall be conducted using the standard measuring equipment to inspect each circuit of the wiring diagram for open or breaks in the circuits to determine compliance with 3.20.3.

4.5.28.3 *Electrical zero.* The course and fine synchros shall be electrically zeroed with respect to each other as specified in the applicable drawings indicated in 3.20.4 and Specification MIL–S–16892.

4.5.29 *Identification of components.* A visual inspection of the serial numbers stenciled on the servo amplifier shall be made to insure that it is the component that has been qualified with the computer as required by 3.21.

5. PREPARATION FOR DELIVERY

5.1. **Packaging, packing and marking.** Packaging, packing and marking shall be in accordance with Specification MIL–P–14232/P8602750. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 **Intended use.** The Ballistics Computer M14A1 with range servo amplifier is used to receive an electrical signal of the log of range data from a range-finder synchro transmitter thereby producing induced voltage at the synchro transformer in the computer. Signal is relayed and amplified in the range servo amplifier to cause operation of the computer. The computer computes superelevation for various ammuni-

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tions and transmits same mechanically and electrically to the other components of the gun sighting system of the 120 MM Gun Tank: M103A2.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number.

(d) Applicable packaging data sheet number (see 5.1).

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